

# Softmax + Cross-Entropy Gradient

## Definitions

$$z_j = \sum_m W_{jm} x_m + b_j, \quad \hat{y}_i = \frac{e^{z_i}}{S}, \quad S = \sum_{k=1}^C e^{z_k}$$
$$L = - \sum_{i=1}^C y_i \ln \hat{y}_i, \quad y \text{ one-hot}, \quad \sum_i y_i = 1$$

The summation indices ( $k, i, m$ ) are dummy indices;  $j$ , the index we differentiate with respect to, is free.

## Intermediate result

$$\ln \hat{y}_i = z_i - \ln S, \quad \frac{\partial S}{\partial z_j} = e^{z_j} \implies \frac{\partial \ln S}{\partial z_j} = \frac{e^{z_j}}{S} = \hat{y}_j$$

## Derivative of $\ln \hat{y}_i$ with respect to $z_j$

$$\frac{\partial \ln \hat{y}_i}{\partial z_j} = \frac{\partial z_i}{\partial z_j} - \frac{\partial \ln S}{\partial z_j} = \delta_{ij} - \hat{y}_j$$

This splits into two cases:

$$\frac{\partial \ln \hat{y}_i}{\partial z_j} = \begin{cases} 1 - \hat{y}_j & (i = j) \\ -\hat{y}_j & (i \neq j) \end{cases}$$

## Applying it to the loss

$$\begin{aligned} \frac{\partial L}{\partial z_j} &= - \sum_i y_i \frac{\partial \ln \hat{y}_i}{\partial z_j} = - \sum_i y_i (\delta_{ij} - \hat{y}_j) \\ &= - \sum_i y_i \delta_{ij} + \hat{y}_j \sum_i y_i = -y_j + \hat{y}_j \quad \left( \sum_i y_i = 1 \right) \end{aligned}$$

$$\boxed{\frac{\partial L}{\partial z_j} = \hat{y}_j - y_j}$$

## Gradient with respect to the parameters

$$\begin{aligned}\frac{\partial z_k}{\partial W_{jm}} &= \delta_{kj} x_m, & \frac{\partial z_k}{\partial b_j} &= \delta_{kj} \\ \frac{\partial L}{\partial W_{jm}} &= \sum_k \frac{\partial L}{\partial z_k} \frac{\partial z_k}{\partial W_{jm}} = (\hat{y}_j - y_j) x_m \\ \frac{\partial L}{\partial b_j} &= \hat{y}_j - y_j\end{aligned}$$

## Batch of $N$ examples

$$\begin{aligned}L &= \frac{1}{N} \sum_{n=1}^N L^{(n)} \\ \frac{\partial L}{\partial W_{jm}} &= \frac{1}{N} \sum_{n=1}^N \left( \hat{y}_j^{(n)} - y_j^{(n)} \right) x_m^{(n)}, & \frac{\partial L}{\partial b_j} &= \frac{1}{N} \sum_{n=1}^N \left( \hat{y}_j^{(n)} - y_j^{(n)} \right)\end{aligned}$$

In matrix form, for  $X \in \mathbb{R}^{N \times M}$  and  $\hat{Y}, Y \in \mathbb{R}^{N \times C}$ :

$$\frac{\partial L}{\partial W} = \frac{1}{N} (\hat{Y} - Y)^\top X, \quad \frac{\partial L}{\partial b} = \frac{1}{N} (\hat{Y} - Y)^\top \mathbf{1}$$